



KRN-EVD-001

Kernellum Evidence Dossier

A claim-to-artifact record for Kernellum Compiler v0.1 and v0.2 alpha.

VERSION	v0.2 / Evidence snapshot	DATE	September 2026
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This dossier collects the public evidence supporting Kernellum's current technical claims. It distinguishes model results, integer-reference behavior, cycle-level behavior, generated RTL simulation, synthesis evidence, and physical-hardware evidence.

The purpose is not to create a second paper. It is to make each claim traceable to a reproducible public artifact and to make missing evidence explicit.

AI-native systems for efficient computing.



Evidence hierarchy

Kernellum treats stronger claims as requiring stronger evidence. A software result is not a hardware timing result; family-mapped synthesis is not place-and-route; modeled latency is not measured latency.

LEVEL	CURRENT STATUS	PUBLIC EVIDENCE
Model / quantization	PASS	manifest.json, compiler.py, TR-001
Cycle model	450/450 exact	cycle_forward implementation and tests
Generated RTL	32/32 golden cases	testbench + golden input/output vectors
Generic synthesis	PASS / 0 CHECK problems	run_eda.sh + BUILD_STATUS.md
ECP5 family synthesis	PASS	run_ecp5.sh; 7,977 LUT4; 8 MULT18X18D
Place-and-route	NOT YET	No named-board P&R; result claimed
Measured FPGA latency/power	NOT YET	No physical-board measurement claimed

01 / WORKLOAD

Demonstrated workload

FIELD	VALUE	BOUNDARY / NOTE
Dataset	scikit-learn handwritten digits	Public demo workload
Network	64 -> 32 -> 16 -> 10	ReLU hidden layers
Held-out samples	450	25% stratified split
Float accuracy	96.22%	Held-out software result
INT8 accuracy	96.44%	Held-out integer pipeline
Float/INT8 agreement	99.78%	Prediction agreement, not accuracy

02 / SEARCH

Architecture search

MAC LANES	CYCLES	MODELED LATENCY @ 100 MHz	MEETS <=10 us
1	2720	27.2 us	no
2	1360	13.6 us	no
4	680	6.8 us	yes - selected
8	340	3.4 us	yes
16	170	1.7 us	yes

The 4-lane design is selected because it is the smallest candidate in the current search set that satisfies the modeled latency target. The 6.8 us value is not a measured FPGA timing result.



03 / TRACEABILITY

Artifact map

CLAIM	ARTIFACT	REPOSITORY PATH / URL
96.44% INT8 accuracy	Manifest	artifacts/digits_int8/manifest.json
450/450 cycle exact	Compiler + tests	kernellum/compiler.py; tests/test_pipeline.py
32/32 RTL golden	Testbench + vectors	artifacts/digits_int8/tb_kernellum_mlp_accel.sv; weights/golden_*.hex
Generated accelerator	SystemVerilog	artifacts/digits_int8/kernellum_mlp_accel.sv
ECP5 mapping	Build status + script	BUILD_STATUS.md; scripts/run_ecp5.sh
Public clean-room rerun	GitHub Actions	.github/workflows/verify.yml
Formal report	TR-001	research/TR-001.md and PDF

04 / SYNTHESIS

Current ECP5 synthesis snapshot

PRIMITIVE	COUNT	INTERPRETATION
LUT4	7,977	Family-mapped synthesis primitive count
MULT18X18D	8	Family-mapped multiplier primitives
TRELLIS_DPR16X4	58	Family-mapped distributed RAM primitives
TRELLIS_FF	145	Family-mapped flip-flop primitives

These counts are not post-place-and-route utilization, do not establish timing closure, and do not establish measured power or latency on a board.

05 / BOUNDARY

Unclaimed evidence

- 01 No named physical FPGA board has yet been validated.
- 02 No nextpnr place-and-route timing result or achieved Fmax is claimed.
- 03 No measured end-to-end board latency is claimed.
- 04 No measured board power or energy per inference is claimed.
- 05 No arbitrary ONNX operator coverage is claimed.
- 06 No ASIC readiness, tapeout readiness, or manufacturable silicon is claimed.

06 / RERUN

Reproduce the public record



```
pip install -e '[test,onnx]' pytest -q python -m kernellum --out artifacts/digits_int8 bash
scripts/run_eda.sh bash scripts/run_ecp5.sh python scripts/build_onnx_demo.py bash
scripts/run_onnx_eda.sh
```

Repository: <https://github.com/sushxnthd/kernellum>